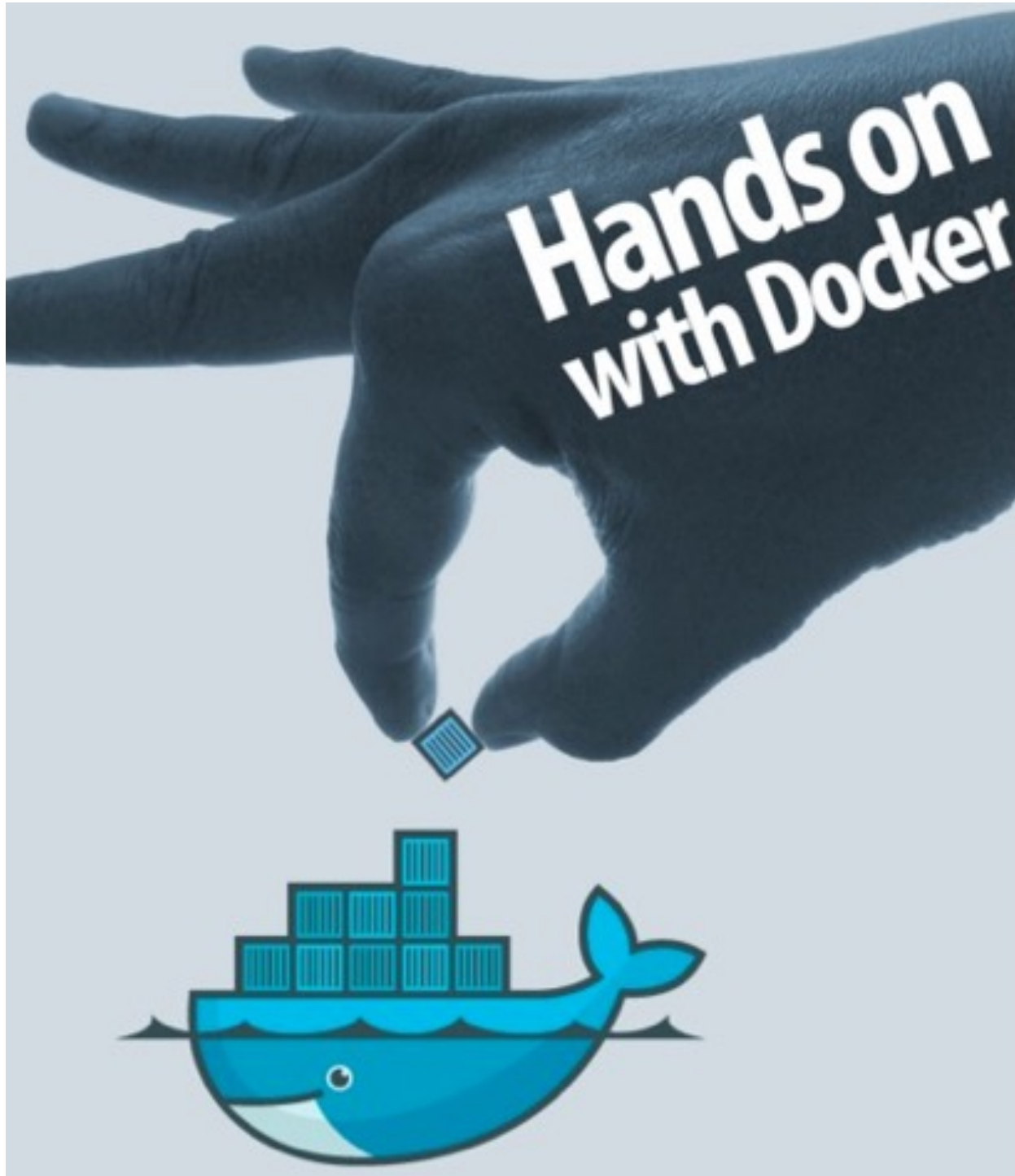




Hands-on

Demo code:

<https://github.com/alwynpan/uom-comp90024/tree/main/docker-demo>

Docker Images

- Public Docker registry (Where you can find the images)
<https://hub.docker.com>

- Docker image name

Syntax: *[registry hostname]/repository[:tag]*

E.g.: *[docker.io/]nginx:alpine* or *nginx[:latest]*

The screenshot shows the Docker Hub page for the `nginx` image. The page header includes the `nginx` logo, the text "Docker Official Image", and download/star counts. A "Tags" tab is selected, showing a list of tags. The first tag, `stable-perl`, is highlighted. Below the tag list, a table shows the digest, OS/ARCH, vulnerabilities, and compressed size for the `stable-perl` tag. The table has four columns: Digest, OS/ARCH, Vulnerabilities, and Compressed size. The first row shows the digest `be1c73bbac7a`, OS/ARCH `linux/386`, vulnerabilities `0 5 0 54 2`, and compressed size `78.31 MB`. The second row shows the digest `96145e50d3ad`, OS/ARCH `linux/amd64`, vulnerabilities `0 5 0 54 2`, and compressed size `79.85 MB`. The third row shows the digest `937d09f15968`, OS/ARCH `linux/arm/v5`, vulnerabilities `0 5 0 54 2`, and compressed size `70.35 MB`. A "+5 more..." link is visible below the table. The page also includes a "docker pull nginx" command and a "Copy" button.

Digest	OS/ARCH	Vulnerabilities	Compressed size
be1c73bbac7a	linux/386	0 5 0 54 2	78.31 MB
96145e50d3ad	linux/amd64	0 5 0 54 2	79.85 MB
937d09f15968	linux/arm/v5	0 5 0 54 2	70.35 MB

Login to a Docker Registry

- Login to a public Docker Registry, *i.e. Docker Hub*.

Syntax: `docker login [OPTIONS] [SERVER]`

E.g.: `docker login --username=foo`

```
> docker login -u alwynpan

Info → A Personal Access Token (PAT) can be used instead.
       To create a PAT, visit https://app.docker.com/settings

Password:
Login Succeeded
```

- Login to a private Docker Registry (e.g., AWS ECR, Nexus Server)

E.g.: `docker login -u AWS https://ecr.ap-southeast-2.amazonaws.com`

```
alwynpan@Alwyns-MBP ~ > docker login -u AWS -p eyJwYXlsb2FkIjoiZC9YQmF5MGNNcXNVekhLejFSUjFpQTJhbnl5Q2Z0LzFUbVVUaXZJN
JzaW9uIjoiMiIsInR5cGUiOiJEQVRBX0tFWSIzImV4cGlyYXRpb24iOiJlNTQ2NjQwMDh9 https://[REDACTED].dkr.ecr.ap-southeast-2.amaz
onaws.com
WARNING! Using --password via the CLI is insecure. Use --password-stdin.
Login Succeeded
```

- Logout

Syntax: `docker logout [SERVER]`

E.g.: `docker logout`

```
> docker logout
Removing login credentials for https://index.docker.io/v1/
```

Pulling a Docker Image

- Pull an image from a public Docker Registry, i.e. Docker Hub.

Syntax: `docker pull NAME[:TAG]`

E.g.: `docker pull nginx`

```
> docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
5716fd893132: Pull complete
0d9786b44f2a: Pull complete
b1a5e9a47bf9: Pull complete
0a43e039475a: Pull complete
881ff00f5a79: Pull complete
f4badebec24: Pull complete
e14d31524fc8: Pull complete
068f82ab28b2: Download complete
6be5707a6ba5: Download complete
Digest: sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest

What's next:
View a summary of image vulnerabilities and recommendations → docker scout quickview nginx
```

- List all images

Syntax: `docker images [OPTIONS] [REPOSITORY[:TAG]]`

E.g.: `docker images` or `docker image ls`

```
alwynpan@Alwyns-MBP ~$ docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
nginx	latest	2bcb04bdb83f	11 days ago	109MB

Pushing a Docker Image

- (Optional) Tag an image

Syntax: `docker tag <SOURCE_IMAGE> <TARGET_IMAGE>`

E.g.: `docker tag nginx alwynpan/comp90024:nginx`

```
alwynpan@Alwyns-MBP ~$ docker tag nginx alwynpan/comp90024:nginx
alwynpan@Alwyns-MBP ~$ docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
alwynpan/comp90024	nginx	2bcb04bdb83f	11 days ago	109MB
nginx	latest	2bcb04bdb83f	11 days ago	109MB

- Push an image

Syntax: `docker push <NAME[:TAG]>`

E.g.: `docker push alwynpan/comp90024:nginx`

```
alwynpan@Alwyns-MBP ~$ docker push alwynpan/comp90024:nginx
The push refers to repository [docker.io/alwynpan/comp90024]
7e274c0effe8: Mounted from library/nginx
dd0338cdfab3: Mounted from library/nginx
5dacd731af1b: Mounted from library/nginx
nginx: digest: sha256:dabec7dece2fff98fb00add2f0b525b7cd4a2cacddcc27ea4a15a7922ea47ea size: 948
```

Run a Docker Container

- (Option 1) Create a container, then start the container

Syntax: `docker create [OPTIONS] IMAGE [COMMAND] [ARG ...]`

E.g.: `docker create --name nginx -p 8080:80 nginx`

Syntax: `docker start [OPTIONS] CONTAINER [CONTAINER ...]`

E.g.: `docker start nginx`

```
alwynpan@Alwyns-MBP ~$ docker create --name nginx -p 8080:80 nginx
51eb524eb9b9bc2a2843bf24d396f6ec5a8c07887e3382e75ab75fef0a135bc3
alwynpan@Alwyns-MBP ~$ docker start nginx
nginx
```

- (Option 2) Run a container

Syntax: `docker run [OPTIONS] IMAGE [COMMAND] [ARG ...]`

E.g.: `docker run --name nginx -p 8080:80 -d nginx`

```
alwynpan@Alwyns-MBP ~$ docker run --name nginx -p 8080:80 -d nginx
54f52f88def92c800cf8194a7557dcbbc6608d1c1365727c3842fe8a688e2d53
```

Run a Docker Container

- (Option 2) Run a container without “-d”

Syntax: *docker run [OPTIONS] IMAGE [COMMAND] [ARG ...]*

E.g.: docker run --name nginx -p 8080:80 nginx

```
> docker run --name nginx -p 8080:80 nginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/04/02 10:10:51 [notice] 1#1: using the "epoll" event method
2026/04/02 10:10:51 [notice] 1#1: nginx/1.29.7
2026/04/02 10:10:51 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/04/02 10:10:51 [notice] 1#1: OS: Linux 6.12.76-linuxkit
2026/04/02 10:10:51 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/04/02 10:10:51 [notice] 1#1: start worker processes
```

```
2026/04/02 10:11:44 [notice] 1#1: signal 17 (SIGCHLD) received from 32
2026/04/02 10:11:44 [notice] 1#1: worker process 32 exited with code 0
2026/04/02 10:11:44 [notice] 1#1: signal 29 (SIGIO) received
2026/04/02 10:11:44 [notice] 1#1: signal 17 (SIGCHLD) received from 30
2026/04/02 10:11:44 [notice] 1#1: worker process 30 exited with code 0
2026/04/02 10:11:44 [notice] 1#1: worker process 31 exited with code 0
2026/04/02 10:11:44 [notice] 1#1: signal 17 (SIGCHLD) received from 31
2026/04/02 10:11:44 [notice] 1#1: worker process 29 exited with code 0
2026/04/02 10:11:44 [notice] 1#1: exit
```

```
> docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
ad07b3dcc020	nginx	"/docker-entrypoint...."	About a minute ago	Exited (0) 42 seconds ago		nginx

List Docker Containers

```
alwynpan@Alwyns-MBP ~ ➤ docker create --name nginx-created -p 8080:80 nginx
34977c499d0694fc1f9e1de732114cd6845be237c56652f1d5426cb67e1d268e
alwynpan@Alwyns-MBP ~ ➤ docker run --name nginx-running -p 8080:80 -d nginx
56aa6c304480dc2674585e27f76a15fdb181518bc027a631d03969c093161543
```

- List running containers
Syntax: docker ps [OPTIONS]
E.g.: docker ps

```
alwynpan@Alwyns-MBP ~ ➤ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
56aa6c304480	nginx	"nginx -g 'daemon of..."	3 seconds ago	Up 2 seconds	0.0.0.0:8080->80/tcp

nginx-running

- List all containers
docker ps -a

```
alwynpan@Alwyns-MBP ~ ➤ docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
56aa6c304480	nginx	"nginx -g 'daemon of..."	About a minute ago	Up About a minute	0.0.0.0:8080->80/tcp
34977c499d06	nginx	"nginx -g 'daemon of..."	About a minute ago	Created	

nginx-running
nginx-created

Stop / Restart / Remove a Docker Container

- Restart a container

Syntax: docker restart [OPTIONS] CONTAINER [CONTAINER ...]

E.g.: docker restart nginx

- Stop the container

Syntax: docker stop [OPTIONS] CONTAINER [CONTAINER ...]

E.g.: docker stop nginx

- Remove a non-running container

Syntax: docker rm [OPTIONS] CONTAINER [CONTAINER ...]

E.g.: docker rm nginx

- Remove a running container

docker rm -f nginx

Running a Shell within a Container

Containers can be started and stopped, and behave like VMs. Instead of using *SSH* to access a VM, a container can be accessed with *exec* command.

(Not recommended, mainly for debug or testing purpose.)

Syntax: docker exec [OPTIONS] CONTAINER COMMAND [ARG ...]

E.g.: docker exec -ti -w /usr/share/nginx/html/ nginx bash

sed -i 's/nginx!/nginx in Docker!/g' index.html

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Welcome to nginx in Docker!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.



Demo 1: Manage data in Docker

- Create a volume

Syntax: docker volume create [OPTIONS] [VOLUME]

E.g.: docker volume create --name htdocs

- Start a container with a **volume** attached

*E.g.: docker run --name nginx-volume -p 8080:80 *
-v htdocs:/usr/share/nginx/html -d nginx

- Start a container with **bind mount** attached

*E.g.: docker run --name nginx-bind -p 8081:80 *
-v \$(pwd)/htdocs:/usr/share/nginx/html -d nginx

With named volume, the content of the container (index.html etc) was there, but with bind mount the directory was empty. Why?

A new named volume's contents can be populated by a container.

What's in a Dockerfile

FROM `nginx:latest` *# Base image, the starting point for your build*

ENV `WELCOME_STRING="nginx in Docker"`

WORKDIR `/usr/share/nginx/html`

COPY `["./entrypoint.sh", "/"]`

RUN `cp index.html index_backup.html; \`
`chmod +x /entrypoint.sh; \`
`apt-get update && apt-get install -qy vim`

ENTRYPOINT `["/entrypoint.sh"]`

CMD `["nginx", "-g", "daemon off;"]`

Run at build time

Run at start up

ENTRYPOINT

- ENTRYPOINT gets executed when the container starts. CMD specifies arguments that will be fed to the ENTRYPOINT.
- Unless it is overridden, ENTRYPOINT will always be executed.

```
#!/usr/bin/env sh
```

```
# Replace the substring with the value of the environment variable ${WELCOME_STRING}
```

```
sed -i 's/Welcome to nginx!/Welcome to "'${WELCOME_STRING}'"/g'  
/usr/share/nginx/html/index.html
```

```
# Make the ENTRYPOINT a pass through, then runs the command specified in CMD. By default, "$@" variable points to the command line arguments.
```

```
exec "$@"
```

```
# ➔ exec nginx -g "daemon off;"
```

Demo 2: Create an image

- Create an image

Syntax: docker build [OPTIONS] PATH

E.g.: docker build -t demo2 .

- Create a container from the image

*docker run --name demo2 -e WELCOME_STRING="COMP90024" \
-p 8081:80 -d demo2*

Welcome to COMP90024!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Scan Docker Image for CVEs

Scan Docker image for CVEs

Syntax: *docker scout cves [OPTIONS] <image name>*

E.g.: *docker scout cves wordpress:6.7.2-php8.1-fpm-alpine*

```
>> docker scout cves wordpress:6.7.2-php8.1-fpm-alpine
✓ SBOM obtained from attestation, 115 packages found
✓ Provenance obtained from attestation
✗ Detected 1 vulnerable package with 1 vulnerability

## Overview
```

	Analyzed Image
Target	wordpress:6.7.2-php8.1-fpm-alpine
digest	a7d2eba0e46c
platform	linux/arm64/v8
provenance	https://github.com/docker-library/wordpress.git
	470084224c4a2b4f0c2e19da2af9ab96298739e4
vulnerabilities	0C 1H 0M 0L
size	95 MB
packages	115
Base image	o1support/staging-arm64v8:9a6fc84dae259d2e09f5470fc7b529e8905ce528913cc082876f1348b4aef4e98382fb9ce49

```
## Packages and Vulnerabilities

0C 1H 0M 0L expat 2.6.4-r0
pkg:apk/alpine/expat@2.6.4-r0?os_name=alpine&os_version=3.21

latest/php8.1/fpm-alpine/Dockerfile.(10:18)
RUN set -eux; \
    apk add --no-cache \
    # in theory, docker-entrypoint.sh is POSIX-compliant, but priority is a working, consistent image
    bash \
    # Ghostscript is required for rendering PDF previews
    ghostscript \
    # Alpine package for "imagemagick" contains ~120 .so files, see: https://github.com/docker-library/wordpress/pull/497
    imagemagick \
    ;

✗ HIGH CVE-2024-8176
https://scout.docker.com/v/CVE-2024-8176
Affected range : <2.7.0-r0
Fixed version  : 2.7.0-r0

1 vulnerability found in 1 package
CRITICAL 0
HIGH 1
MEDIUM 0
LOW 0

What's next:
View base image update recommendations → docker scout recommendations wordpress:6.7.2-php8.1-fpm-alpine
```


Docker Scout Recommendations

Get Docker base image update recommendations

Syntax: *docker scout recommendations [OPTIONS] <image name>*

E.g.: *docker scout recommendations wordpress:6.7.2-php8.1-fpm-alpine*

```
> docker scout recommendations wordpress:6.7.2-php8.1-fpm-alpine
✓ SBOM of image already cached, 115 packages indexed
```

Target	wordpress:6.7.2-php8.1-fpm-alpine
digest	a7d2eba0e46c
Base image	oisupport/staging-arm64v8:9a6fc84daae259d2e09f5470fc7b529e8905ce528913cc082876f1348b4aef4e

No recommendations

Demo 3: WordPress + MySQL + phpMyAdmin

WordPress is a free and open-source content management system based on PHP and MySQL.

Demo 3: use [Compose](#) file to start a WordPress website including WordPress, MySQL and phpMyAdmin.

- Start the containers
[Syntax](#): docker compose up [OPTIONS]
E.g.: docker compose up -d
- Stop the containers
[Syntax](#): docker compose stop [OPTIONS] [SERVICE ...]
E.g.: docker compose stop
- Remove the containers
[Syntax](#): docker compose down [OPTIONS]
E.g.: docker compose down

Demo 4: Docker Hardening

- Update the Dockerfile from Demo 2 to use a non-root user
 - Not be able to install any packages
 - Not be able to modify any files owned by other users
 - Not be able to create any file outside the directories owned by nginx user

```
nginx@690a3f58a6c1:/usr/share/nginx$ touch demo
touch: cannot touch 'demo': Permission denied
nginx@690a3f58a6c1:/usr/share/nginx$ apt update
Reading package lists... Done
E: List directory /var/lib/apt/lists/partial is missing. - Acquire (13: Permission denied)
nginx@690a3f58a6c1:/usr/share/nginx$ apt install vim
E: Could not open lock file /var/lib/dpkg/lock-frontent - open (13: Permission denied)
E: Unable to acquire the dpkg frontend lock (/var/lib/dpkg/lock-frontent), are you root?
nginx@690a3f58a6c1:/usr/share/nginx$ rm /etc/issue
rm: remove write-protected regular file '/etc/issue'? yes
rm: cannot remove '/etc/issue': Permission denied
nginx@690a3f58a6c1:/usr/share/nginx$
```

- Run the rootless Nginx image with a read-only filesystem
 - Not able to write any file to the file system even if obtain root access

```
nginx@9ddb97a12f01:/docker-entrypoint.d$ touch demo
touch: cannot touch 'demo': Read-only file system
```

Demo 4: Docker Debug

- Use docker debug command to access the container

Syntax: `docker debug [OPTIONS] <container name>`

E.g.: `docker debug demo4-1`

```
root@690a3f58a6c1 /usr/share/nginx/html [demo4-1]
docker > apt update
Hit:1 http://deb.debian.org/debian bookworm InRelease
Hit:2 http://deb.debian.org/debian bookworm-updates InRelease
Hit:3 http://deb.debian.org/debian-security bookworm-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@690a3f58a6c1 /usr/share/nginx/html [demo4-1]
docker > █
```

- Install packages in read-only containers

```
root@9ddb97a12f01 /usr/share/nginx/html [demo4-2]
docker > install vim
Tip: You can install any package available at: https://search.nixos.org/packages.
trace: warning: cudaPackages.autoAddDriverRunpath is deprecated, use pkgs.autoAddDriverRunpath instead
trace: warning: cudaPackages.autoAddOpenGLRunpathHook is deprecated, use pkgs.autoAddDriverRunpathHook instead
trace: warning: cudaPackages.autoFixElfFiles is deprecated, use pkgs.autoFixElfFiles instead
trace: warning: cudaPackages.autoAddDriverRunpath is deprecated, use pkgs.autoAddDriverRunpath instead
trace: warning: cudaPackages.autoAddOpenGLRunpathHook is deprecated, use pkgs.autoAddDriverRunpathHook instead
trace: warning: cudaPackages.autoFixElfFiles is deprecated, use pkgs.autoFixElfFiles instead
trace: warning: cudaPackages.autoAddDriverRunpath is deprecated, use pkgs.autoAddDriverRunpath instead
trace: warning: cudaPackages.autoAddOpenGLRunpathHook is deprecated, use pkgs.autoAddDriverRunpathHook instead
trace: warning: cudaPackages.autoFixElfFiles is deprecated, use pkgs.autoFixElfFiles instead
trace: warning: eww now can build for X11 and wayland simultaneously, so 'eww-wayland' is deprecated, use the normal 'eww' package instead.
trace: warning: lxd has been renamed to lxd-lts
trace: warning: lxd-unwrapped has been renamed to lxd-unwrapped-lts
trace: warning: nixfmt was renamed to nixfmt-classic. The nixfmt attribute may be used for the new RFC 166-style formatter in the future, which is currently available as nixfmt-rfc-style
trace: warning: nvtop has been renamed to nvtopPackages.full
trace: warning: nvtop-amd has been renamed to nvtopPackages.amd
trace: warning: nvtop-intel has been renamed to nvtopPackages.intel
trace: warning: nvtop-msm has been renamed to nvtopPackages.msm
trace: warning: nvtop-nvidia has been renamed to nvtopPackages.nvidia
trace: warning: writeReferencesToFile is deprecated in favour of writeClosure
replacing old 'vim-9.1.0377'
installing 'vim-9.1.0377'
```